Module 5: Deep Learning and Computer Vision

1. Implement a simple feedforward neural network using NumPy.
2. Implement a custom activation function and integrate it into a neural network.
3. Use TensorFlow to create a neural network model for the MNIST dataset and evaluate its accuracy.
4. Implement forward propagation in a neural network manually.
5. Write a function to compute the loss for a given dataset and weights.
6. Create a Keras model to classify the Fashion MNIST dataset and evaluate its accuracy.
7. Build a CNN for classifying cats vs. dogs and analyze the misclassified examples.